

Hybrid Handcrafted and CNN Feature Approach



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Skin Lesion Classification for Monkeypox Detection

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Outline

Introduction
Dataset & Preprocessing
Methodology

Results
Discussion
Conclusion



Background

Monkeypox (MPX) is a zoonotic disease caused by the Monkeypox virus.

- Historically endemic to Central and West Africa
- 2022–2023 global outbreak highlighted urgent need for rapid diagnostics
- Visual inspection of skin lesions is the primary diagnostic method
- Easily confused with chickenpox, measles, and other skin rashes

Challenge: Automated classification is needed for timely diagnosis and containment.



Sample skin lesion images

Problem Statement & Objectives

Problem Statement

Skin lesions from Monkeypox are visually similar to other conditions, making manual diagnosis error-prone and time-consuming.

Research Objectives

1. Implement robust lesion segmentation from skin background
2. Extract rich handcrafted features (color, texture, shape, edges, frequency)
3. Leverage transfer-learned CNN embeddings (ResNet18)
4. Compare handcrafted, CNN, and hybrid feature representations
5. Evaluate four classifiers for binary MPX vs. non-MPX classification

Dataset Description

Source: Monkeypox Skin Lesion Dataset (Kaggle)

- 228 skin lesion images (224×224 RGB)
- **Monkeypox (MPX):** 102 images
- **Non-Monkeypox (ETC):** 126 images
- Slightly imbalanced: $\sim 45\%$ MPX, 55% ETC

Split: Stratified 70/30

- Training: 160 images
- Testing: 68 images



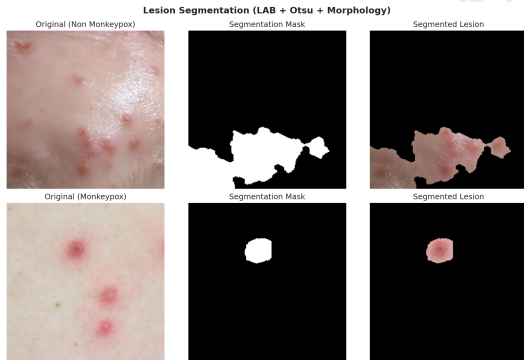
Dataset sample images

Lesion Segmentation Pipeline

Method: LAB + CLAHE + Otsu + Morphology

1. Convert BGR $\rightarrow$ LAB color space
2. Apply CLAHE to L^* and a^* channels
3. Combine L^* and inverted a^* (weights 0.3, 0.7)
4. Otsu thresholding for binarization
5. Morphological closing + opening (5×5 ellipse)
6. Keep largest connected component

Result: Clean binary mask isolating the primary lesion.



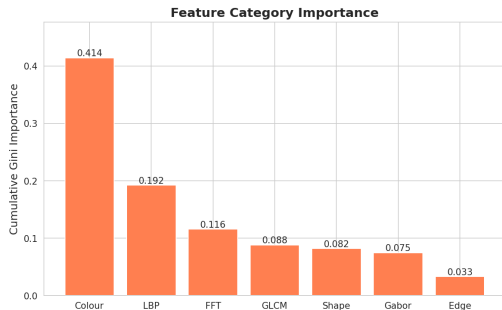
Segmentation results

Handcrafted Feature Extraction

Seven categories, ~200 raw features

- **Color:** Moments + HSV histograms + percentiles in BGR/HSV/LAB (84)
- **GLCM:** Contrast, dissimilarity, homogeneity, energy, correlation (10)
- **LBP:** Uniform patterns, radius=2, P=16 (18)
- **Gabor:** 3 orientations $\times$ 2 scales, mean+std (12)
- **Shape:** Hu moments + geometric properties (12)
- **Edge:** Canny density + Sobel gradient stats (3)
- **FFT:** Radial power spectrum rings (16)

Feature importance (by category)



Cumulative Gini importance

CNN Feature Extraction

Transfer Learning with ResNet18

- Pre-trained on ImageNet (1.2M images, 1000 classes)
- Remove final FC layer
- Extract 512-dim embedding from global average pooling
- Inference mode (no gradient computation)

Preprocessing

- Resize to 224×224
- ImageNet normalization
- $\mu = [0.485, 0.456, 0.406]$
- $\sigma = [0.229, 0.224, 0.225]$

Feature Representations

1. **Handcrafted only** (155 dims)
2. **CNN only** (512 dims)
3. **Hybrid (full)** (667 dims)
4. **Hybrid (selected)** (200 dims via MI)

All features Z-score normalized before classification.

Classification Algorithms

Classifier	Key Hyperparameters
SVM (RBF)	$C = 10, \gamma = \text{scale}$
Random Forest	200 estimators, max depth=10
XGBoost	200 estimators, max depth=6, lr=0.1
Logistic Regression	$C = 1.0$, max iter=5000

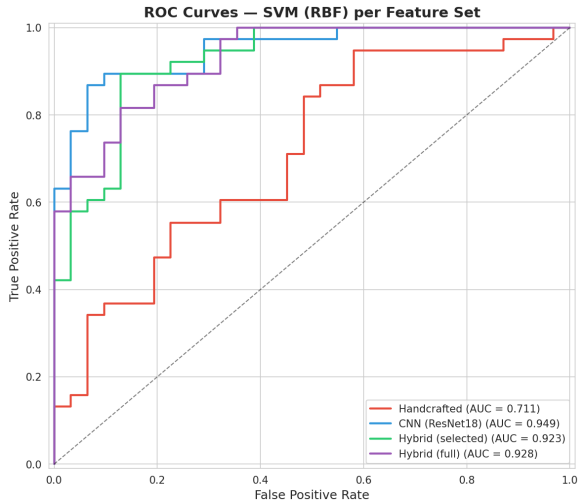
- 5-fold stratified cross-validation on training set
- Fixed hyperparameters across all feature sets for fair comparison
- Independent test set evaluation

Classification Results

Feature Set	Classifier	Acc.	F1	Test AUC	CV AUC
CNN (ResNet18)	SVM (RBF)	0.855	0.853	0.949	0.891
Hybrid (full)	XGBoost	0.841	0.838	0.890	0.798
Hybrid (selected)	XGBoost	0.855	0.848	0.907	0.833
Hybrid (selected)	SVM (RBF)	0.841	0.840	0.923	0.916
CNN (ResNet18)	Random Forest	0.812	0.806	0.911	0.821
CNN (ResNet18)	Logistic Reg.	0.826	0.823	0.887	0.847
Hybrid (full)	SVM (RBF)	0.826	0.825	0.928	0.915
Handcrafted	XGBoost	0.754	0.752	0.840	0.768
Handcrafted	SVM (RBF)	0.594	0.592	0.711	0.776

Best result: CNN-only + SVM (RBF) achieves **85.5% accuracy** and **AUC = 0.949**

ROC Curves

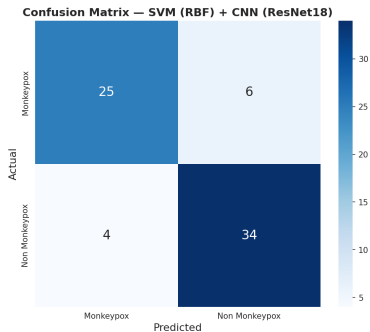


Observations

- CNN-only dominates all other feature sets
- AUC = 0.949 (near-perfect discrimination)
- Hybrid approaches competitive but not superior
- Handcrafted-only significantly lower performance

ROC curves for SVM (RBF)

Confusion Matrix & Classification Report



Confusion matrix (best model)

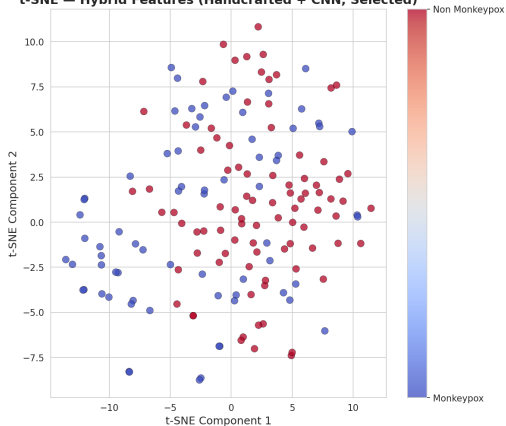
Classification Report (SVM + CNN-only)

Class	Precision	Recall	F1
Monkeypox	0.86	0.81	0.83
Non-Monkeypox	0.85	0.89	0.87
Macro Avg	0.86	0.85	0.85

- Correctly identifies 81% of MPX cases
- Correctly identifies 89% of non-MPX cases
- Balanced misclassifications

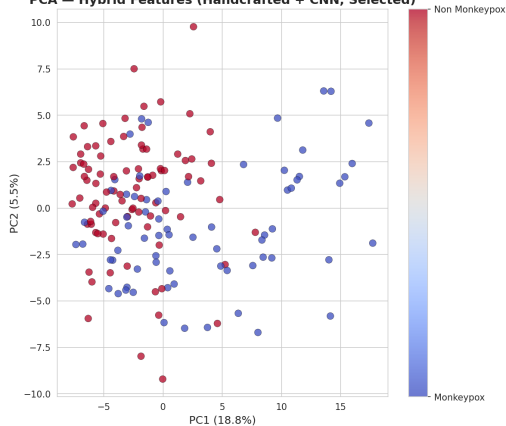
Feature Space Visualization

t-SNE — Hybrid Features (Handcrafted + CNN, Selected)



t-SNE (hybrid selected)

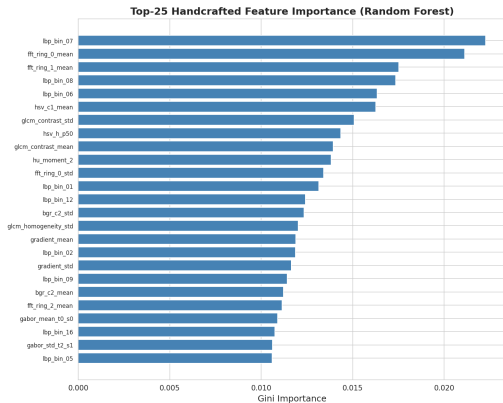
PCA — Hybrid Features (Handcrafted + CNN, Selected)



PCA (hybrid selected)

Classes show moderate separability; some overlap remains consistent with challenging visual diagnosis

Feature Importance Analysis



Top-25 handcrafted features (RF)

Key Findings

- **Color features** most important (HSV/LAB moments)
- **GLCM texture** second most discriminative
- Gabor and edge features also contribute
- Aligns with dermatological knowledge:
 - MPX lesions have characteristic red-purple hues
 - Specific surface textures (umbilicated vesicles)

Discussion

CNN Features

- ResNet18 embeddings highly effective even **without fine-tuning**
- Capture high-level patterns: shape, color, texture
- Best performance: AUC = 0.949

Handcrafted Features

- Moderate performance (best AUC = 0.840)
- Color and GLCM most discriminative
- Interpretable and lightweight

Hybrid Approach

- Did **not** surpass CNN-only baseline
- Small dataset (228 images) causes overfitting with 667 dims
- Feature selection (200 dims) helps but still below CNN-only
- May outperform with larger dataset

Classifier Choice

SVM (RBF) consistently best across all feature sets due to ability to model non-linear decision boundaries in high-

Conclusion & Future Work

Summary of Contributions

1. Robust LAB-based lesion segmentation pipeline
2. Comprehensive handcrafted feature extraction (7 categories)
3. Transfer-learned CNN embeddings from ResNet18
4. Systematic comparison: handcrafted vs. CNN vs. hybrid
5. Best result: **85.5% accuracy, AUC = 0.949**

Future Directions

- Collect larger, multi-center dataset for fine-tuning
- Integrate attention mechanisms for region highlighting
- Deploy as lightweight mobile app for point-of-care use
- Investigate ensemble methods combining multiple CNN architectures

References

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Thank You

Questions?

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